

Operator 4: Algebraic Construction of Local-Topology Degree-Statistic Operator $\mathcal{M}_{\text{degree}}$ and Rigorous Positive-Definite Boundedness Proof for Dirichlet Energy Functional

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Resource-Availability Statement This framework is built upon Status-Relational Entropy (SRE) Dynamics. All theoretical materials are archived in the Zenodo open-access repository. **This manuscript suite, including system papers, application developments, scientific hypotheses, full algebraic derivations for operators 1-6 and simulation code, is fully open-source.** Operators 7, 8, 9, 10 are subsequent closed-source commercial core modules and are not part of this manuscript suite.

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According to the SRE principle, the physical foundation originates from information statistics.

Document positioning: This manuscript constitutes core component of Phase-1 Homogeneous-Metric Operator Suite within Status-Relational-Entropy (SRE) Dynamics. Operator 4 receives intermediate outputs from Operator 1 and Operator 2, and supplies spectral-prior parameters for downstream Operator 5. It belongs to the open-source Operator 1-6 suite; Operators 7-10 are closed-source commercial-core modules and are outside the scope of this document.

Abstract

This paper rigorously derives and constructs algebraically the **Local-Topology Degree-Statistic Operator ($\mathcal{M}_{\text{degree}}$, Operator 4)**, a core member of the Phase 1 homogeneous-metric operator suite under Status-Relational-Entropy (SRE) Dynamics. Targeting discontinuous step-noise induced by distributed-actor local-horizon fragmentation and local zero-degree vacuum singularities, this operator builds analytic homogeneous smoothing measures by fusing the 2-Step Graph-Walk Kernel and spectral-bound regularisation terms.

The highest-priority mathematical achievement of this work is the complete proof of the **Rigid-Clamping Theorem for the Lower Bound of the Dirichlet-Energy Functional**. Under extremely-sparse or zero-degree-vacuum scenarios, Operator 4 autonomously activates the global algebraic-connectivity scale-adjustment valve, rigidly clamping the global-graph Dirichlet energy functional inside a fully

positive-definite compact subspace:

$$\mathcal{E}_D(E_s) \geq \lambda_2(n) \cdot \|E_s\|_2^2 > 0.$$

Fundamentally it eliminates logarithmic-divergence singularities originating from floating-point round-off errors of distributed cut-sets, guaranteeing mathematical completeness for long-timescale distributed-engineering deployment.

1. Introduction & Physical Philosophy

During outward-expansion cascades of SRE-dynamics networks, direct adoption of raw discrete node-degrees as topological-evolution probability measures inevitably encounters two major mathematical-physical difficulties:

1. **Localized Horizon Fragmentation:** Distributed actors can only observe local-fragmented topology, causing discontinuity step-jumps of global-connectivity measures along boundaries and introducing high-frequency step-noise.
2. **Zero-Degree Vacuum Singularity:** At distributed-cut-set edges of sparse graphs, vacuum states with local node in/out-degree equal to zero readily emerge. Conventional graph-dynamics suffer algebraic-degeneracy under such conditions, triggering floating-point round-off-error divergences of logarithmic or fractional terms within governing equations.

To overcome these defects, the design of Operator 4 fully implements the physical-philosophy of **Spectral Homogeneous Smoothing**. It completely abandons metrics relying on extrinsic background spacetime or manually hard-coded prescriptions. Instead, endogenous damping is composed from the system's own **2-Step Graph-Walk Kernel invariants** together with global spectral priors, endowing discrete network-graph topological-evolution with smooth continuum rheological properties.

2. Algebraic Spaces & Notation Conventions

For rigid interfacing with subsequent operators within the pipeline, mathematical spaces and symbolic matrices acted-upon by Operator 4 are defined below:

- **Local-Fragment Binary-Spin Symmetric Square Matrix (M_Ω):** Read-only persistent matrix passed and finalised by Operator 1 and Operator 2, defined over compact discrete domain:

$$M_\Omega \in \mathcal{M}_{\text{spin}}^{(N_K)} \subseteq \{-1, +1\}^{N_K \times N_K}$$

It satisfies strict symmetry $M_{\Omega,ij} = M_{\Omega,ji}$ and contains no zero entries.

- **1-Based Index Baseline Set ($\mathcal{J}_{N_K}$):** Matrix slicing and element access strictly follow $\mathcal{J}_{N_K} = \{1, 2, \dots, N_K\}$.
- **Forward-Measure Degree Mapping (τ):** Since entries inside M_Ω are strictly ± 1 , the baseline for forward-degree extraction operator is defined as element-absolute-value summation. Owing to symmetry, local out-degree cardinality D_{ii}^{out} and in-degree cardinality D_{ii}^{in} degenerate completely to equivalence:

$$D_{ii}^{\text{out}} \equiv \sum_{j=1}^{N_K} |M_{\Omega,ij}| = D_{ii}^{\text{in}}$$

- **Diagonal Self-Loop Term (D_{ii}^{self}):** Vertex self-spin feature mapped onto matrix main diagonal, satisfying:

$$D_{ii}^{\text{self}} \equiv M_{\Omega,ii} \in \{-1, +1\}$$

- **Global Algebraic-Connectivity Fiedler Prior ($\lambda_2(n)$):** Second-smallest eigenvalue of global Laplacian matrix at previous pulse step, streamed out via low-rank iteration by upstream Operator 6. Thanks to causal-safety interception from Operator 2, the full-network graph always remains fully connected, hence rigidly locked: $\lambda_2(n) > 0$.
- **Global Spectral-Radius Prior (α_n):** Maximum eigenvalue $\lambda_{\max}(n)$ of the global Laplacian matrix.
- **Independent Spectral-Boundary Regularisation Invariant ($\epsilon_{\text{topo}}^{(s)}$):** Protective damping attached onto local main-diagonal entries, solved analytically from Spectral-Area-Ratio; it always satisfies $\epsilon_{\text{topo}}^{(s)} \in \mathbb{R}^+$.
- **Graph-Map Output Edge-Weight ($W_{e(i,j)}^{(s-1)}$):** Homogeneous-smoothed continuous scalar weight output by Operator 4, constituting core parameter for flow-field divergence control.

3. Analytical Derivation of Operator 4 Standard Algebraic Equations

Analytic construction for homogeneous weight coefficient $W_{e(i,j)}^{(s-1)}$ is compound-superimposed in two parts: **normalised topological base component** and **graph-walk-kernel logarithmic damping component**.

3.1 Local-Normalised Cross-Correlation Base Component

To mitigate scaling-inflation effects from distributed expansion of full-network node count n , geometric mean of local two-vertex degrees $\sqrt{D_{ii}^{\text{out}} \cdot D_{jj}^{\text{in}}}$ must be introduced. Meanwhile, to counter zero-degree vacuum inside the denominator, composite coherent counter-balancing is performed using global algebraic-connectivity, diagonal self-loop polarities and independent spectral-boundary regulariser, constructing base-term with strict lower-bound protection:

$$W_{\text{base}}(i, j) \equiv \frac{\sqrt{D_{ii}^{\text{out}} \cdot D_{jj}^{\text{in}}}}{\sqrt{\lambda_2(n) + D_{ii}^{\text{self}} + D_{jj}^{\text{self}} + \epsilon_{\text{topo}}^{(s)}}}$$

3.2 2-Step-Graph-Walk-Kernel Logarithmic-Perturbation Damping Component

According to algebraic-graph-theory principles, absolute value $|(M_{\Omega}^2)_{ij}|$ of multiplied inner-product over fragmented spin-square matrix perfectly characterises total number of coherent interference and destructive-cancellation events for all formal causal paths of length-2 travelling from node i to node j . Non-linear logarithm compression must be adopted to suppress long-range cascade divergence.

Define global-evolution-scale-regulating factor as $\frac{\lambda_2(n)}{\alpha_n}$. Using local-energy-shunting term $\alpha_n + \sqrt{D_{ii}^{\text{out}} \cdot D_{jj}^{\text{in}}}$ for perturbation amortised compensation, logarithmic-perturbation-damping formula is derived:

$$W_{\text{damp}}(i, j) \equiv 1 + \frac{|(M_{\Omega}^2)_{ij}| \cdot \ln \left(1 + \frac{\lambda_2(n)}{\alpha_n} \right)}{\alpha_n + \sqrt{D_{ii}^{\text{out}} \cdot D_{jj}^{\text{in}}}}$$

3.3 Complete Closed-Form General Equation of Operator 4

Homogeneously cascade-multiply geometric-normalised measure of base-component and non-linear causal-perturbation of damping-component, finally establishing the standard explicit algebraic equation for Operator 4 Local-Topology Degree-Statistic:

$$W_{e(i,j)}^{(s-1)} \equiv \frac{\sqrt{D_{ii}^{\text{out}} \cdot D_{jj}^{\text{in}}}}{\sqrt{\lambda_2(n) + D_{ii}^{\text{self}} + D_{jj}^{\text{self}} + \epsilon_{\text{topo}}^{(s)}}} \cdot \left(1 + \frac{|(M_{\Omega}^2)_{ij}| \cdot \ln \left(1 + \frac{\lambda_2(n)}{\alpha_n} \right)}{\alpha_n + \sqrt{D_{ii}^{\text{out}} \cdot D_{jj}^{\text{in}}}} \right)$$

4. Rigorous Mathematical Proof: Rigid-Clamping Theorem for Lower Bound of Dirichlet-Energy Functional

Theorem 4.1 (Theorem of Complete Positive-Definiteness of Dirichlet-Energy and Spontaneous Singularity Elimination)

Under extremely-sparse distributed-evolution or zero-degree-vacuum-edge scenarios (namely local-degree statistics facing limit collapse $D_{ii}^{\text{out}} \rightarrow 0$, $D_{jj}^{\text{in}} \rightarrow 0$, driving output-weight towards zero convergence $W_e \rightarrow 0$), Operator 4 autonomously activates the global algebraic-connectivity scale-adjustment valve, rigidly clamping the global-graph Dirichlet-Energy-Functional algebraic lower-bound inside fully positive-definite compact subspace:

$$\mathcal{E}_D(E_s) \geq \lambda_2(n) \cdot \|E_s\|_2^2 > 0$$

Thereby fundamentally eliminating singularity hazards of logarithmic-computation divergences caused by floating-point round-off errors at algebraic bottom-layer.

Proof Steps

Step 1: Singularity and Boundedness Analysis under Zero-Degree Limit

Suppose system evolves towards an extremely-sparse distributed-sub-domain cut-set, target nodes i and j behave as fully-isolated within current fragment, i.e. their degree-cardinalities approach zero-vacuum state:

$$D_{ii}^{\text{out}} \rightarrow 0, \quad D_{jj}^{\text{in}} \rightarrow 0$$

Now inspect base-term $W_{\text{base}}(i, j)$ inside Operator 4 closed-form equation. Its numerator deterministically converges to zero via geometric-mean effect:

$$\lim_{D \rightarrow 0} \sqrt{D_{ii}^{\text{out}} \cdot D_{jj}^{\text{in}}} = 0$$

Next decompose its denominator term. Subject to strict binary-spin constraints upon M_Ω , vertex self-loop features on main-diagonal satisfy extremum bounds:

$$D_{ii}^{\text{self}} \in \{-1, +1\} \implies D_{ii}^{\text{self}} + D_{jj}^{\text{self}} \geq -2$$

Owing to global Fiedler algebraic-connectivity prior $\lambda_2(n) > 0$ and independent spectral-boundary regulariser $\epsilon_{\text{topo}}^{(s)} > 0$, denominator term is rigidly locked outside negative-value domain:

$$\sqrt{\lambda_2(n) + D_{ii}^{\text{self}} + D_{jj}^{\text{self}} + \epsilon_{\text{topo}}^{(s)}} \geq \sqrt{\lambda_2(n) - 2 + \epsilon_{\text{topo}}^{(s)}} > 0$$

Since denominator possesses strictly-positive non-zero real lower-bound, while numerator converges to zero, base-term unconditionally monotonically converges towards zero:

$$\lim_{D \rightarrow 0} W_{\text{base}}(i, j) = 0$$

Now inspect logarithmic-perturbation damping-term $W_{\text{damp}}(i, j)$. As M_Ω becomes fully-sparse adjacency under this condition, its 2-step-walk product magnitude $|(M_\Omega^2)_{ij}| \rightarrow 0$, driving damping-term to converge towards multiplicative-identity element 1:

$$\lim_{D \rightarrow 0} W_{\text{damp}}(i, j) = 1 + \frac{0 \cdot \ln(1 + \cdot)}{\alpha_n + 0} = 1$$

Compound limit-values of base-term and damping-term directly prove output-weight range-boundary of Operator 4 under zero-degree-vacuum scenario:

$$\lim_{D \rightarrow 0} W_{e(i,j)}^{(s-1)} = 0 \times 1 \equiv 0$$

This boundary demonstrates: Operator 4 completely cuts-off physical coherence between zero-degree nodes, without generating undefined floating-point singularities such as $\frac{0}{0}$ or $\ln(0)$.

Step 2: Operator-Algebra Mapping for Dirichlet-Energy Functional

Let high-dimensional directed co-variant error-flow-field vector at current refresh micro-step be $E_s \in \mathbb{R}^n$. Corresponding global-graph Dirichlet-Energy-Functional $\mathcal{E}_D(E_s)$ is strictly expressed by Laplacian quadratic-form:

$$\mathcal{E}_D(E_s) \equiv E_s^T L_G E_s = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n W_{e(i,j)}^{(s-1)} (E_s(i) - E_s(j))^2$$

When large-scale topological collapse happens on distributed cut-sets and massive edge-weights $W_e \rightarrow 0$, global-graph Laplacian matrix L_G faces systematic risk of large-area eigenvalue-degeneration towards zero, further triggering collapse of entire energy-functional space.

Step 3: Variational Projection and Courant-Fischer Theorem Clamping

To prevent energy-space collapse, perform orthogonal-cohomology decomposition for error-flow-field vector E_s . Project it into complement-space of Laplacian constant-kernel subspace, stripping translation-invariance and enforcing full-network sum-conservation gauge constraint:

$$\mathbf{1}^T E_s = \sum_{i=1}^n E_s(i) = 0$$

Within directed chain-complex space, for arbitrary non-zero co-variant vector $E_s \neq 0$ algebraically orthogonal against all-ones constant vector $\mathbf{1}$, by the **Courant-Fischer Min-Max Variational Principle**, variational lower-bound of its Rayleigh quotient is uniquely locked by second-smallest eigenvalue (algebraic-connectivity):

$$\lambda_2(n) \equiv \min_{E_s \perp \mathbf{1}, E_s \neq 0} \frac{E_s^T L_G E_s}{\|E_s\|_2^2}$$

Hence rigid-inequality bound for global-graph Dirichlet-Energy quadratic-form can be derived:

$$\mathcal{E}_D(E_s) = E_s^T L_G E_s \geq \lambda_2(n) \cdot \|E_s\|_2^2$$

Step 4: Complete Positive-Definite Closure under Cohomology-Causality Interception

Per Version 18.0 cascade-timing-dependency specification, upstream **Operator 2 (Pre-emptive Cohomological Random-Pruning Operator $\mathcal{O}_{\text{gate_batch}}$)** acts as causal-safety interceptor. At pre-processing phase it performs pre-emptive batch exemption for non-isomorphic bridge-edge combinations which may induce spanning-tree degeneracy and physical splitting of full-network graph; these critical channels are rigidly forced permanently conductive ($\chi_e \leftarrow 1$).

Through underlying geometric-safety interception of Operator 2, topological-space connectivity of mother-graph obtains absolute rigid safeguard. By fundamental algebraic-graph-theory theorem, necessary-and-sufficient condition for full-graph connectivity is its Laplacian Fiedler eigenvalue strictly greater than zero. Therefore global algebraic-connectivity passed-from-prior satisfies hard positive-definite red-line:

$$\lambda_2(n) > 0$$

Given non-zero flow-field, its squared L_2 -norm term must be strictly positive: $\|E_s\|_2^2 > 0$. Substitute this red-line into variational inequality, completing closed-proof of continuous inequality-chain:

$$\mathcal{E}_D(E_s) \geq \lambda_2(n) \cdot \|E_s\|_2^2 > 0$$

Conclusion of Proof

Even under distributed-iteration inside local-actor nodes, where floating-point round-off errors induce extreme fluctuations of arguments inside operator-endogenous logarithmic terms, the rigid lower-bound $\lambda_2(n) > 0$ of global algebraic-connectivity firmly locks functional-space of full-graph inside fully-positive-definite compact subspace.

Dirichlet-Energy-Functional cannot collapse down to zero. Denominator-terms inside logarithmic expressions obtain globally-safe field-energy support; singularity hazards originating from floating-point round-off divergences are spontaneously eliminated.

■ Theorem 4.1 is complete.

5. Conclusions and Next-Step Pipeline Roadmap

Operator 4 ($\mathcal{M}_{\text{degree}}$) achieves smooth homogeneous-metric on purely topological dimension via cascade-composition of local-normalised cross-correlation kernel and 2-Step-Graph-Walk causal kernel. The successful proof for Rigid-Clamping Theorem of Dirichlet-Energy lower-bound establishes deadlock-free mathematical foundation for long-timescale stable distributed-simulation of entire SRE-dynamics network.

Following Version 18.0 dependency-topology pipeline order, after mathematical derivations of Operator 4 are fully validated, work proceeds immediately onto the second core component within **Phase 1**

Homogeneous-Metric Operator Suite:

- **Advance to Operator 6 (Subspace-Spectral-Sieve & Splicing Operator $\mathcal{P}_{\text{sieve}} \cup \mathcal{O}_{\text{splice}}$):** Derive **Rayleigh-Ritz algebraic boundary-splicing kernel**. Adaptive Lanczos low-rank iteration completely abolishes full-network heavy global eigendecomposition, supplying streamed-prior solutions of $\lambda_2(n)$ and α_n for Operator 4, crushing overall computational complexity from $\mathcal{O}(n^3)$ down into sparse-local bound $\mathcal{O}(m_g \cdot k_{\text{rank}})$.